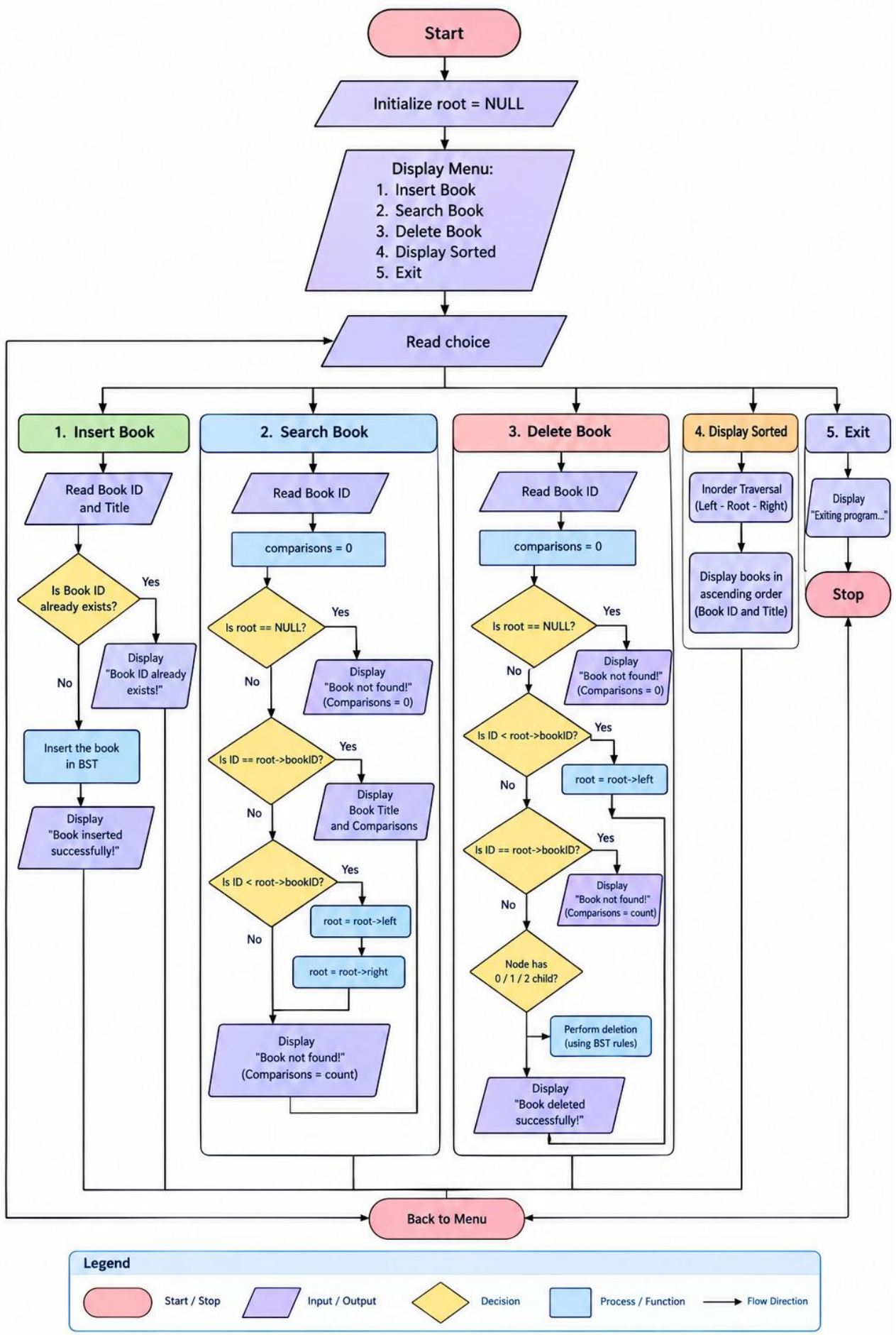
Flowchart


Start
Initialize root = NULL
Display Menu:
1. Insert Book
2. Search Book
3. Delete Book
4. Display Sorted
5. Exit
Read choice
1. Insert Book
Read Book ID and Title
Is Book ID already exists?
Yes
Display "Book ID already exists!"
No
Insert the book in BST
Display "Book inserted successfully!"
2. Search Book
Read Book ID
comparisons = 0
Is root == NULL?
Yes
Display "Book not found!" (Comparisons = 0)
No
Is ID == root->bookID?
Yes
Display Book Title and Comparisons
No
Is ID < root->bookID?
Yes
root = root->left
No
root = root->right
Display "Book not found!" (Comparisons = count)
3. Delete Book
Read Book ID
comparisons = 0
Is root == NULL?
Yes
Display "Book not found!" (Comparisons = 0)
No
Is ID < root->bookID?
Yes
root = root->left
No
Is ID == root->bookID?
Yes
Display "Book not found!" (Comparisons = count)
No
Node has 0 / 1 / 2 child?
Perform deletion (using BST rules)
Display "Book deleted successfully!"
4. Display Sorted
Inorder Traversal (Left - Root - Right)
Display books in ascending order (Book ID and Title)
5. Exit
Display "Exiting program..."
Stop
Back to Menu
Legend
Start / Stop
Input / Output
Decision
Process / Function
Flow Direction